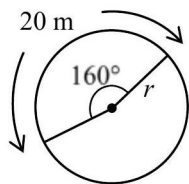


mb-pc40s-2019-jan

Typst conversion review. OCR content and mathematical accuracy require editorial review.

Question 1

Determine the length of the radius, r , given an arc length of 20 metres and a central angle of 160° .



Solution

$$\theta = (160) \left(\frac{\pi}{180} \right)$$

$$= \frac{8\pi}{9}$$

$$r = \frac{s}{\theta}$$

$$r = \frac{20}{\left(\frac{8\pi}{9} \right)}$$

$$r = \frac{180}{8\pi} \text{ m}$$

1 mark for conversion

1 mark for substitution

2 marks

or

$$r = 7.162 \text{ m}$$

Question 2

(4hash 1

There are eight cars parked in a row. Determine the number of possible arrangements of these eight cars if Mrs. Jones must always park in the third spot and Mr. Rodriguez must always park in the last spot.

Solution

$$6 \cdot 5 \cdot 1 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 1$$

Question 3

Bill wins \$1300000 in a lottery and invests the entire amount at an annual interest rate of 2.5% compounded quarterly. He will withdraw \$10000 at the end of every three months.

Determine, algebraically, the total number of withdrawals, including the partial amount, that Bill can make until there is no money left. Express your answer as a whole number.

Use the formula:

$$PV = \frac{R[1 - (1 + i)^{-n}]}{i}$$

where

PV = the present value deposited

R = the amount of each withdrawal

n = the number of equal withdrawals

$i = \frac{\text{the annual interest rate (in decimal form)}}{\text{the number of compounding periods}}$

Solution

$$1300000 = \frac{10000 \left[1 - \left(1 + \frac{0.025}{4} \right)^{-n} \right]}{\frac{0.025}{4}}$$

$$8125 = 10000[1 - (1.00625)^{-n}]$$

$$0.8125 = 1 - (1.00625)^{-n}$$

$$-0.1875 = -(1.00625)^{-n}$$

$$\log(0.1875) = -n \log 1.00625$$

$$-\frac{\log(0.1875)}{\log(1.00625)} = n$$

$$268.672348 = n$$

Bill can make 269 withdrawals.

$\frac{1}{2}$ mark for substitution

$\frac{1}{2}$ mark for simplification

$\frac{1}{2}$ mark for applying logarithms

1 mark for power law

$\frac{1}{2}$ mark for solving for n

3 marks

Question 4

(Mill)

Determine and simplify the 12th term in the binomial expansion of $\left(x^3 - \frac{1}{2x^2}\right)^{12}$.

Solution

$$t_{12} = {}_{12}C_{11}(x^3)^1\left(-\frac{1}{2x^2}\right)^{11}$$

2 marks (1 mark for ${}_{12}C_{11}$; $\frac{1}{2}$ mark for each consistent factor)

$$t_{12} = (12)(x^3)\left(-\frac{1}{2048x^{22}}\right)$$

$$t_{12} = -\frac{12}{2048x^{19}}$$

or

1 mark for simplification ($\frac{1}{2}$ mark for coefficient; $\frac{1}{2}$ mark for exponent)

3 marks

$$t_{12} = -\frac{3}{512x^{19}}$$

or

$$t_{12} = -0.006x^{-19}$$

Question 5

(Mili)

Solve, algebraically.

$$e^{2x-3} = 7^{x+1}$$

Solution

$$\ln e^{2x-3} = \ln 7^{x+1}$$

$$(2x - 3) \ln e = (x + 1) \ln 7$$

$$2x - 3 = x \ln 7 + \ln 7$$

$$2x - x \ln 7 = \ln 7 + 3$$

$$x(2 - \ln 7) = \ln 7 + 3$$

$$x = \frac{\ln 7 + 3}{2 - \ln 7}$$

$$x = 91.438783$$

$$x = 91.439$$

$\frac{1}{2}$ mark for applying logarithms

1 mark for power law

$\frac{1}{2}$ mark for collecting terms with x

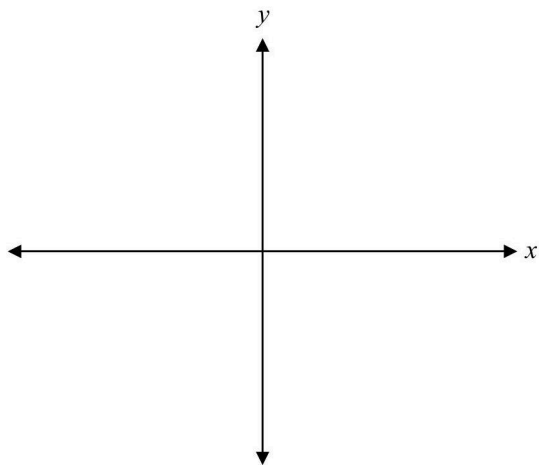
$\frac{1}{2}$ mark for isolating x

$\frac{1}{2}$ mark for evaluating quotient of logarithms

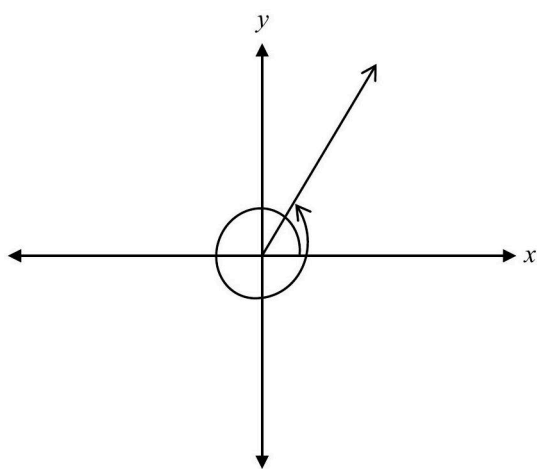
3 marks

Question 6

Sketch the angle $\frac{7\pi}{3}$ in standard position.



Solution



$\frac{1}{2}$ mark for an appropriate angle in quadrant I

$\frac{1}{2}$ mark for correct number of revolutions

1 mark

Note:

- If the directional arrow is not indicated, deduct an E1 error (final answer not stated).

Question 7

Determine, algebraically, all of the zeros of the polynomial function $P(x) = x^4 - 5x^3 - 4x^2 + 20x$.

Solution

$$P(x) = x(x^3 - 5x^2 - 4x + 20)$$

$$P(2) = 2(2^3 - 5(2)^2 - 4(2) + 20)$$

$$P(2) = 0$$

$$\therefore (x - 2) \text{ is a factor}$$

1 mark for identifying one possible value of x

2	1	-5	-4	20
	↓	2	-6	-20
	1	-3	-10	0

1 mark for synthetic division

(or any other equivalent strategy)

$$P(x) = x(x - 2)(x^2 - 3x - 10)$$

$$0 = x(x - 2)(x - 5)(x + 2)$$

$$x = 0, x = 2, x = 5, x = -2$$

$\frac{1}{2}$ mark for identifying all the factors

$\frac{1}{2}$ mark for consistent zeros

3 marks

Question 8

Justify why four of the terms in the binomial expansion of $(-x + y)^6$ are positive.

Solution

$$(-x)^6(y)^0, (-x)^5(y)^1, (-x)^4(y)^2, \dots \\ x^6, -x^5y, x^4y^2, \dots$$

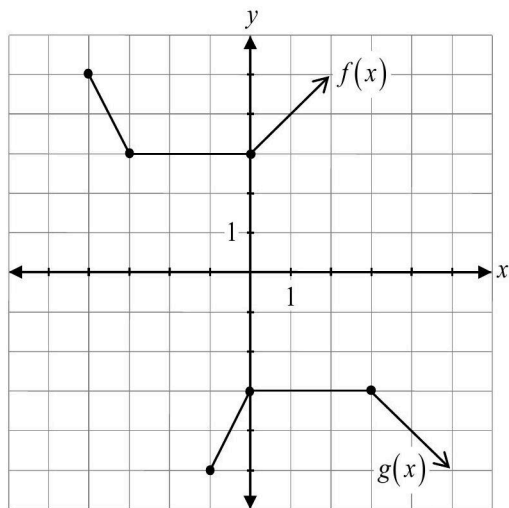
There are 7 terms. The first term is positive and the signs alternate.

1 mark for justification

1 mark

Question 9

Determine the equation of $g(x)$ in terms of $f(x)$.



$g(x) =$

Solution

$g(x) = -f(x - 3)$ 1 mark for vertical reflection
1 mark for horizontal translation

Question 10

Prove the identity for all permissible values of x .

$$\frac{\sin x + \tan x}{\cot x + \csc x} = \frac{\sin^2 x}{\cos x}$$

Left-Hand Side	Right-Hand Side

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{\sin x + \frac{\sin x}{\cos x}}{\frac{\cos x}{\sin x} + \frac{1}{\sin x}}$ $\frac{\frac{\sin x \cos x + \sin x}{\cos x}}{\frac{\cos x + 1}{\sin x}}$ $\frac{\sin x(\cos x + 1)}{\cos x} \cdot \frac{\sin x}{(\cos x + 1)}$ $\frac{\sin^2 x}{\cos x}$	$\frac{\sin^2 x}{\cos x}$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Method 2

Left-Hand Side	Right-Hand Side
$\frac{\sin x + \tan x}{\frac{1}{\tan x} + \frac{1}{\sin x}}$ $\frac{\sin x + \tan x}{\frac{\sin x + \tan x}{\sin x \tan x}}$	$\frac{\sin^2 x}{\cos x}$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 11

Given that the point $(-2, 1)$ is on the graph of $y = f(x)$, describe how the coordinates of the corresponding point on the graph of $y = f(4x)$ are different.

Solution

The x -value is divided by 4 .

1 mark

Question 12

Using the laws of logarithms, completely expand the expression:

$$\log\left(\frac{x^2\sqrt{y}}{w-1}\right)$$

Solution

$$2\log x + \frac{1}{2}\log y - \log(w-1)$$

1 mark for product law

1 mark for power law ($\frac{1}{2}$ mark for each)

1 mark for quotient law

3 marks

Question 13

Given $\sec \theta = -\frac{5}{4}$ and $\tan \theta > 0$, state the quadrant in which θ terminates.

Justify your answer.

Solution

Since $\sec \theta$ is negative in quadrants II and III, and $\tan \theta$ is positive in quadrants I and III,

θ terminates in quadrant III.

1 mark for justification

1 mark

Question 14

State an equation of a rational function that has a vertical asymptote at $x = -8$ and a horizontal asymptote at $y = 9$.

Solution

$$y = \frac{9x}{x+8}$$

1 mark for vertical asymptote

1 mark for horizontal asymptote

2 marks

- Other equations are possible.

Question 15

Given the point $(5, 1)$, state the coordinates of the corresponding point after a reflection across the line $y = x$.

Solution

$(1, 5)$

1 mark

Question 16

Simplify.

$$\frac{(n-13)!}{(n-12)!}$$

Solution

$$\frac{(n-13)!}{(n-12) \frac{(n-13)!}{1}} = \frac{1}{n-12}$$

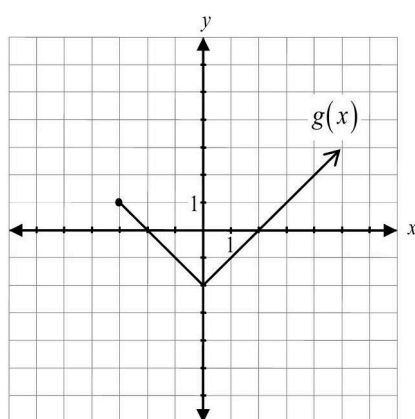
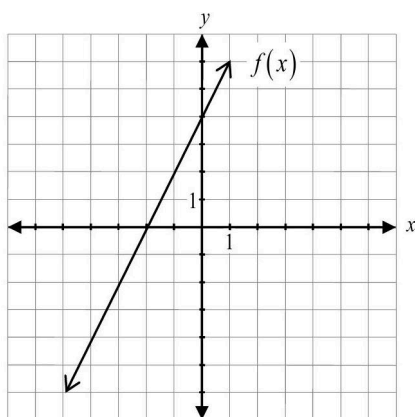
1 mark for factorial expansion

1 mark

Question 17

a) 1 mark b) 1 mark

Given the graphs of $y = f(x)$ and $y = g(x)$,



a) determine the value of $f(g(2))$.

b) determine the value of $(g - f)(-3)$.

Solution

$$g(2) = 0$$

$\frac{1}{2}$ mark for the value of $g(2)$

$$f(0) = 2$$

$\frac{1}{2}$ mark for the value of $f(g(2))$ consistent with $g(2)$

1 mark

b) determine the value of $(g - f)(-3)$.

$$\begin{aligned} g(-3) - f(-3) \\ 1 - (-2) \end{aligned}$$

$\frac{1}{2}$ mark for the values of $g(-3)$ and $f(-3)$

3

$\frac{1}{2}$ mark for value of $(g - f)(-3)$ consistent with $g(-3)$

and $f(-3)$

1 mark

Question 18

Identify the remainder when $P(x) = 3x^3 - x^2 + 1$ is divided by $(x - 2)$.

A.

-27

B.

-19

C.

11

D.

21

Solution

Correct answer: D

21

Question 19

Identify the logarithmic form of $2^x = \frac{1}{4}$.

A.

$$\log_2 x = \frac{1}{4}$$

B.

$$\log_x 2 = \frac{1}{4}$$

C.

$$\log_2\left(\frac{1}{4}\right) = x$$

D.

$$\log_x\left(\frac{1}{4}\right) = 2$$

Solution

Correct answer: C

$$\log_2\left(\frac{1}{4}\right) = x$$

Question 20

Leah's Pizzeria offers 9 different pizza toppings. Identify the expression that represents the number of different types of pizzas, with 3 different toppings, that can be made.

A.

$${}_9C_3$$

B.

$${}_9P_3$$

C.

$$\frac{9!}{3!}$$

D.

$$9!3!$$

Solution

Correct answer: A

$${}_9C_3$$

Question 21

Given $(5, -4)$ is a point on the graph of $y = f(x)$, identify the corresponding point on the graph of $y = \frac{1}{f(x)}$.

A.

$$\left(\frac{1}{5}, -4\right)$$

B.

$$\left(5, -\frac{1}{4}\right)$$

C.

$$\left(\frac{1}{5}, -\frac{1}{4}\right)$$

D.

$$(-4, 5)$$

Solution

Correct answer: B

$$\left(5, -\frac{1}{4}\right)$$

Question 22

Identify the non-permissible value of x for $1 + \sec x$ over $[0, \pi]$.

A.

$$0$$

B.

$$\frac{\pi}{4}$$

C.

$$\frac{\pi}{2}$$

D.

$$\pi$$

Solution

Correct answer: C

$$\frac{\pi}{2}$$

Question 23

Indicate the combination that represents the circled term in the given row of Pascal's triangle.

1

46

(4) 1

A.

${}_4C_3$

B.

${}_4C_4$

C.

${}_5C_3$

D.

${}_5C_4$

Solution

Correct answer: A

${}_4C_3$

Question 24

Identify the x -intercept on the graph of $f(x) = \sqrt{2(x+5)}$.

A.

−5

B.

0

C.

$\sqrt{10}$

D.

5

Solution

Correct answer: A

−5

Question 25

Identify the coterminal angle of $\frac{\pi}{5}$ over the interval $-\pi \leq \theta \leq 4\pi$.

A.

$$-\frac{9\pi}{5}$$

B.

$$-\frac{\pi}{5}$$

C.

$$\frac{3\pi}{5}$$

D.

$$\frac{11\pi}{5}$$

Solution

Correct answer: D

$$\frac{11\pi}{5}$$

Question 26

Given $f(x) = \{(2, 6), (3, 2), (3, 4), (6, 5)\}$, identify the value of $f(f(2))$.

A.

3

B.

4

C.

5

D.

6

Solution

Correct answer: C

5

Question 27

The graph of $f(x) = (x - 1)^2$ is translated 2 units to the left and 3 units up. Identify the equation of the transformed graph, $g(x)$.

A.

$$g(x) = (x + 1)^2 + 3$$

B.

$$g(x) = (x - 3)^2 + 3$$

C.

$$g(x) = (x + 2)^2 + 3$$

D.

$$g(x) = (x - 2)^2 + 3$$

Solution

Correct answer: A

$$g(x) = (x + 1)^2 + 3$$

Question 28

Given $\csc \theta = -\frac{8}{5}$, determine the exact value of $\cos 2\theta$.

Solution

$$\begin{aligned}\cos 2\theta &= 1 - 2\sin^2 \theta \\ &= 1 - 2\left(-\frac{5}{8}\right)^2 \\ &= \frac{14}{64}\end{aligned}$$

1 mark for the value of $\sin \theta$

1 mark for substitution into correct identity

2 marks

$$\begin{aligned}&\text{or} \\ &= \frac{7}{32}\end{aligned}$$

Question 29

Determine the period of the sinusoidal function, $f(x) = -6 \cos\left(\frac{\pi}{6}(x + 1)\right) + 5$.

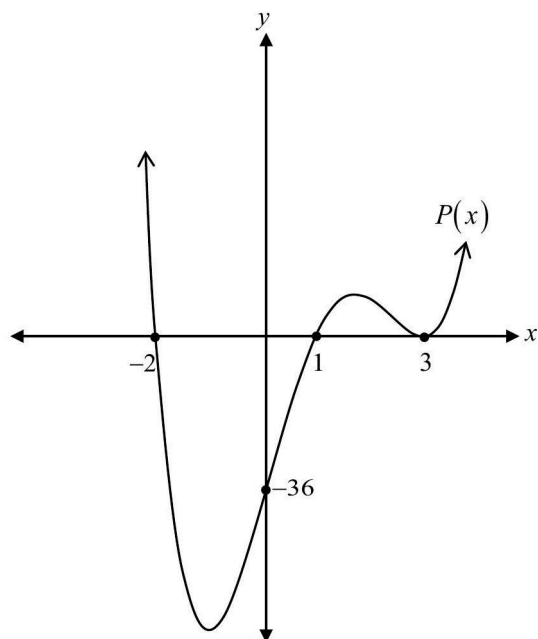
Solution

$$\begin{aligned}\text{Period} &= \frac{2\pi}{\left(\frac{\pi}{6}\right)} \\ &= 12\end{aligned}$$

1 mark

Question 30

Determine, algebraically, the equation of $P(x)$, given the graph of the polynomial function $P(x)$.



$P(x) =$

—

Solution

$$P(x) = a(x - 3)^2(x - 1)(x + 2)$$

$$-36 = a(-3)^2(-1)(2)$$

$$-36 = -18a$$

$$a = 2$$

$$P(x) = 2(x - 3)^2(x - 1)(x + 2)$$

$\frac{1}{2}$ mark for factors of $P(x)$

$\frac{1}{2}$ mark for multiplicity of 2 at $x = 3$

$\frac{1}{2}$ mark for substitution of $P(0) = -36$

$\frac{1}{2}$ mark for correct value of a

2 marks

Question 31

Solve $2 \sin^2 \theta - 7 \sin \theta - 4 = 0$ where $\theta \in \mathbb{R}$.

Solution

$$(2 \sin \theta + 1)(\sin \theta - 4) = 0$$

$$\sin \theta = -\frac{1}{2}$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\theta = \frac{7\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

$$\theta = \frac{11\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

$$\sin \theta = 4$$

1 mark for solving for $\sin \theta$ ($\frac{1}{2}$ mark for each branch)

1 marks for solving for θ (1 mark for each branch)

4 marks

Question 32

Justify that the shapes of the graphs of $f(x) = (x + 1)^2(x - 1)$ and $g(x) = (x + 1)^2(x - 1)^3$ are different as they approach the x -intercept at $x = 1$.

Solution

At $x = 1$, both graphs pass through the x -axis, however, the graph of $g(x)$ flattens as it passes through the x -axis.

Question 33

Determine the exact value of $\cot \theta$ if $\cos \theta = -\frac{4}{7}$ and $\sin \theta$ is positive.

Solution

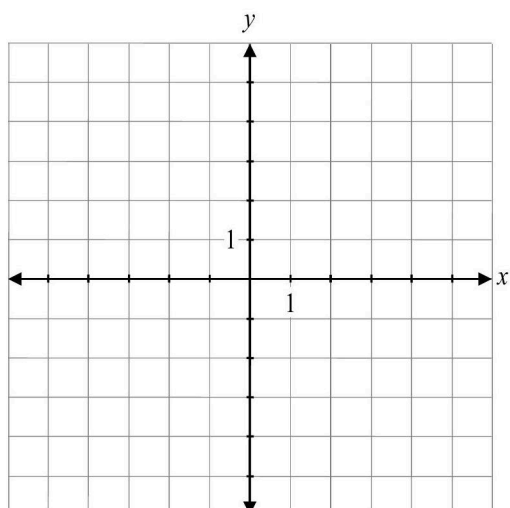
$$\begin{array}{ll} x^2 + y^2 = r^2 & \\ (-4)^2 + y^2 = (7)^2 & 1/2 \text{ mark for substitution} \\ y^2 = 49 - 16 & \\ y = \pm\sqrt{33} & 1/2 \text{ mark for solving for } y \\ \cot \theta = \frac{-4}{\sqrt{33}} & 1 \text{ mark for consistent value of } \cot \theta \left(\frac{1}{2} \text{ mark for quadrant; } 1/2 \text{ mark for value} \right) \\ \text{or} & 2 \text{ marks} \\ \cot \theta = \frac{-4\sqrt{33}}{33} & \end{array}$$

Note:

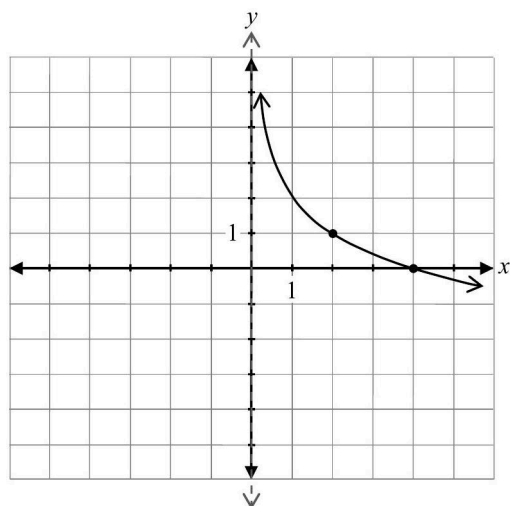
- Accept any of the following values for y : $y = \pm\sqrt{33}$, $y = \sqrt{33}$, or $y = -\sqrt{33}$.

Question 34

Sketch the graph of $f(x) = -\log_2(x) + 2$.



Solution



1 mark for asymptotic behaviour approaching $x = 0$

1 mark for vertical reflection

1 mark for vertical translation

3 marks

Question 35

State the range of $f(x) = \sqrt{x+4}$.

Range:

Solution

Range: $[0, \infty)$

1 mark

or

Range: $\{y \in \mathbb{R} \mid y \geq 0\}$

Question 36

Sophie correctly solved the logarithmic equation, $\log_7(x - 1) = \log_7(2x - 2)$.

$$\begin{aligned}x - 1 &= 2x - 2 \\ -1 + 2 &= 2x - x\end{aligned}$$

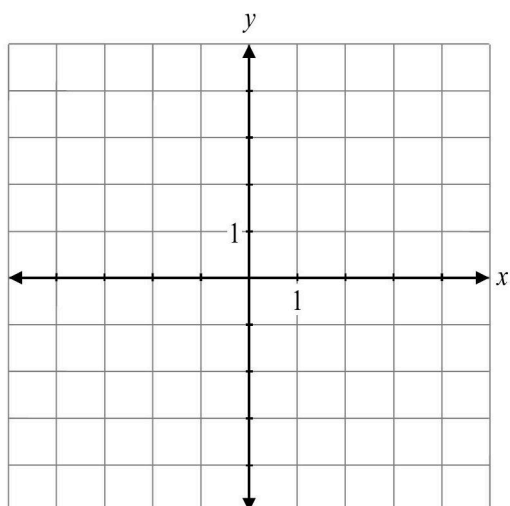
Explain why $x = 1$ is an extraneous root.

Solution

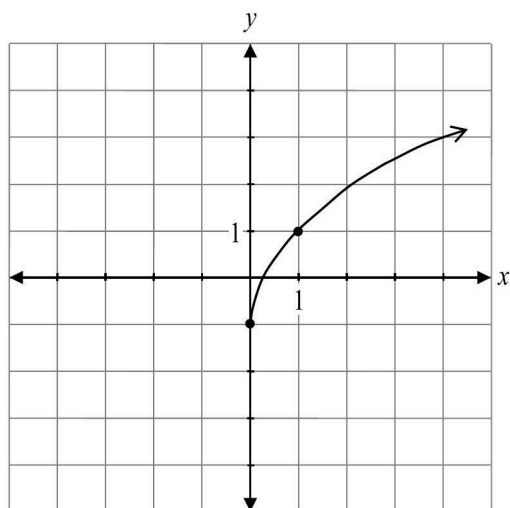
The root, $x = 1$, is an extraneous root because the argument of a logarithm cannot be zero.

Question 37

Sketch the graph of $f(x) = \sqrt{4x} - 1$.



Solution



1 mark for shape of a radical function

1 mark for horizontal compression

1 mark for vertical translation

3 marks

Question 38

Solve, algebraically.

$${}_nC_2 = 2n + 7$$

Solution

$$\frac{n!}{(n-2)!2!} = 2n + 7$$

$$\frac{n(n-1)(n-2)!}{(n-2)!} = 2!(2n+7)$$

$$n(n-1) = 2(2n+7)$$

$$n^2 - n = 4n + 14$$

$$n^2 - 5n - 14 = 0$$

$$(n+2)(n-7) = 0$$

$$n \geq -2 \quad n = 7$$

$\frac{1}{2}$ mark for simplification

$\frac{1}{2}$ mark for the permissible value of n

$\frac{1}{2}$ mark for showing the rejection of the extraneous root

Question 39

Given $f(x) = x^2 - 1$ and $g(x) = x - 3$, explain why the domain of $h(x) = \frac{f(x)}{g(x)}$ has a restriction when $x = 3$.

Solution

When $x = 3$, the denominator is equal to zero and it is not possible to divide by zero.

1 mark

Question 40

Evaluate.

$$\frac{\cot\left(\frac{11\pi}{6}\right)\sin\left(-\frac{4\pi}{3}\right)}{\cos\left(\frac{2\pi}{3}\right)}$$

Solution

$(-\sqrt{3})(\sqrt{3})$ 1 mark for $\cot\left(\frac{11\pi}{6}\right)$ ($\frac{1}{2}$ mark for quadrant; $\frac{1}{2}$ mark for value)

1 mark for $\sin\left(-\frac{4\pi}{3}\right)$ ($\frac{1}{2}$ mark for quadrant; $\frac{1}{2}$ mark for value)

1 mark for $\cos\left(\frac{2\pi}{3}\right)$ ($\frac{1}{2}$ mark for quadrant; $\frac{1}{2}$ mark for value)

$$\left(-\frac{3}{2}\right)\left(-\frac{2}{1}\right)$$

3 marks

3

Question 41

Solve, algebraically.

$$\log_2(\log_3 x) = 2$$

Solution

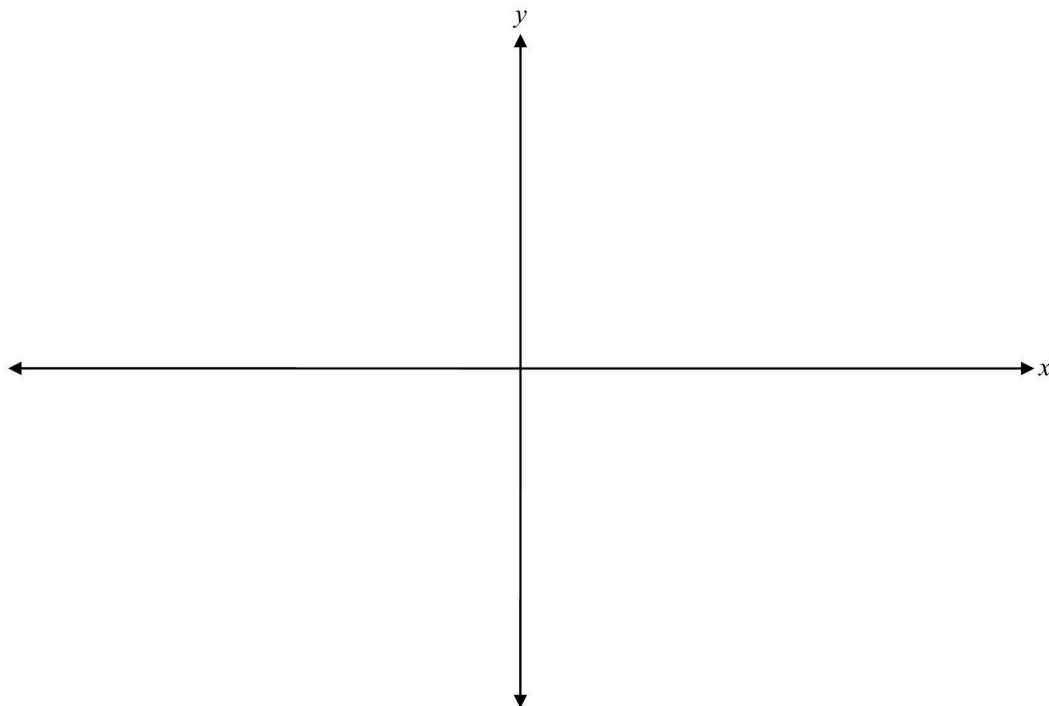
$$2^2 = \log_3 x \quad \text{mark for exponential form}$$

$$4 = \log_3 x \quad \text{mark for exponential form}$$

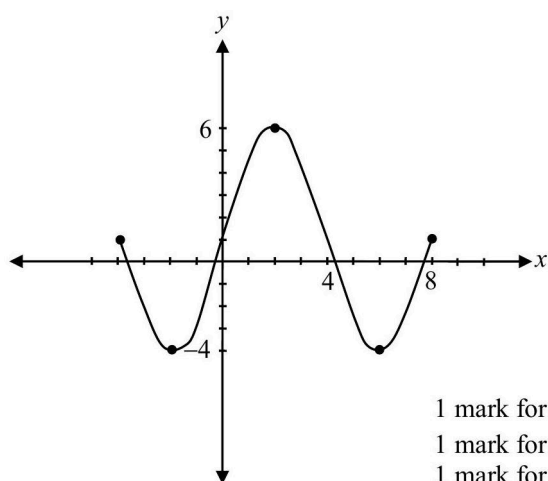
$$x = 3^4 \quad \text{1 mark}$$

Question 42

Sketch the graph of the function $y = 5 \sin\left(\frac{\pi}{4}x\right) + 1$ over the domain $[-4, 8]$.



Solution

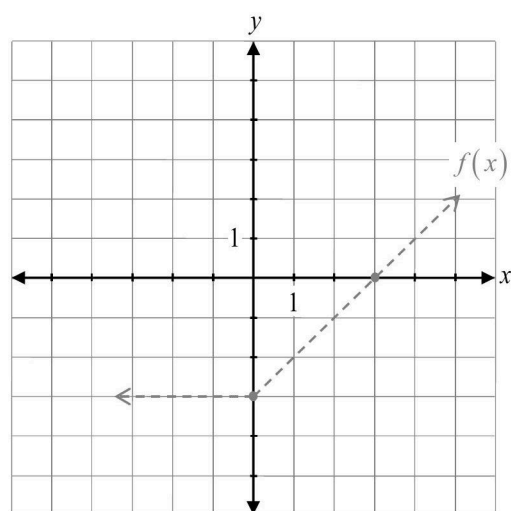
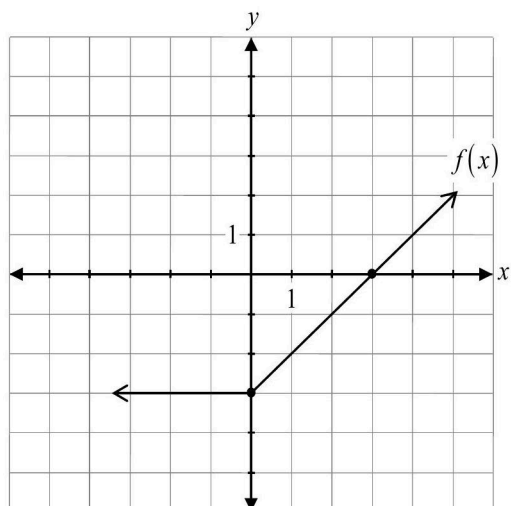


- 1 mark for shape of $y = \sin x$
- 1 mark for amplitude
- 1 mark for period
- 1 mark for vertical translation

4 marks

Question 43

Given the graph of $y = f(x)$, sketch the graph of $y = |f(-x)|$.



The graph of

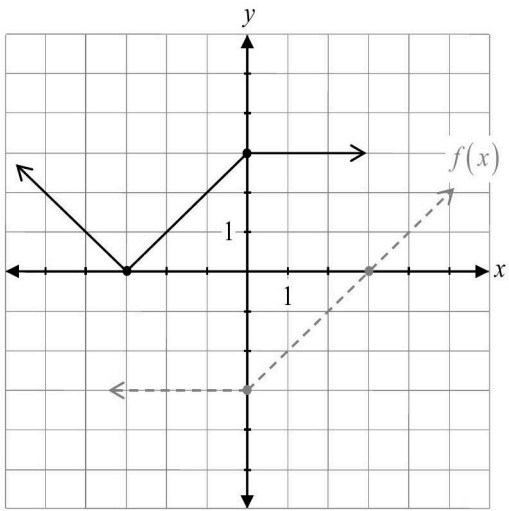


$f(x)$ has already
been drawn for
your reference.

No marks will be
awarded for the
graph of $f(x)$.

Solution

Solution



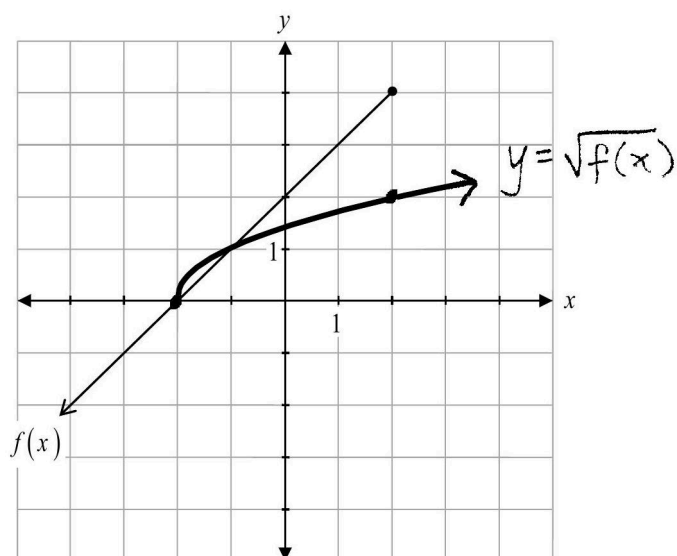
1 mark for horizontal reflection

1 mark for absolute value

2 marks

Question 44

Savannah used the graph of $y = f(x)$ to sketch the graph of $y = \sqrt{f(x)}$. Her solution is given below. Describe her error.



Solution

Savannah did not restrict the domain at $x = 2$.

Question 45

Determine the exact value of $\sin\left(\frac{13\pi}{12}\right)$.

Solution

$$\begin{aligned}\sin\left(\frac{3\pi}{4} + \frac{\pi}{3}\right) &= \sin\left(\frac{3\pi}{4}\right) \cos\left(\frac{\pi}{3}\right) + \cos\left(\frac{3\pi}{4}\right) \sin\left(\frac{\pi}{3}\right) \\ &= \left(\frac{\sqrt{2}}{2}\right) \left(\frac{1}{2}\right) + \left(-\frac{\sqrt{2}}{2}\right) \left(\frac{\sqrt{3}}{2}\right) \\ &= \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4} \\ &= \frac{\sqrt{2} - \sqrt{6}}{4}\end{aligned}$$

1 mark for substitution into correct identity

2 marks ($\frac{1}{2}$ mark for each exact value)

3 marks

Note:

- Other combinations are possible.

Question 46

State the equation of the horizontal asymptote of $f(x) = \frac{3x}{x-1}$.

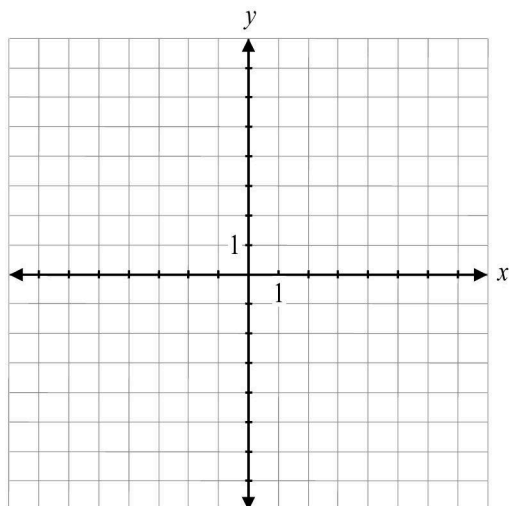
Solution

$$y = 3$$

1 mark

Question 47

Sketch the graph of $f(x) = \frac{5x-10}{x^2+x-6}$.



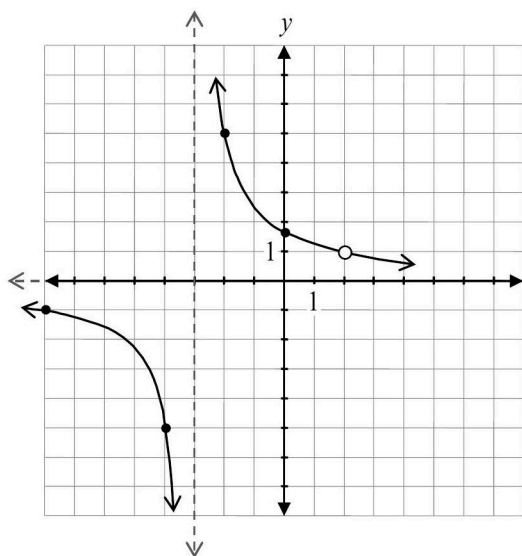
Solution

$$\begin{aligned} f(x) &= \frac{5x-10}{x^2+x-6} \\ &= \frac{5(x-2)}{(x-2)(x+3)} \\ &= \frac{5}{x+3}, x \neq 2 \end{aligned}$$

$\therefore$ there is a point of discontinuity (hole) at $(2, 1)$

vertical asymptote at $x = -3$

horizontal asymptote at $y = 0$



1 mark for asymptotic behaviour approaching $x = -3$

1 mark for asymptotic behaviour approaching $y = 0$

1 mark for a point of discontinuity (hole) at $(2, 1)$

$\frac{1}{2}$ mark for graph left of $x = -3$

$\frac{1}{2}$ mark for graph right of $x = -3$

4 marks

Question 48

Determine, algebraically, the inverse of $f(x) = 3x + 4$.

Solution

Let $f(x) = y$

$$y = 3x + 4$$

$$x = 3y + 4 \quad 1 \text{ mark for switching } x \text{ and } y \text{-values}$$

$$x - 4 = 3y$$

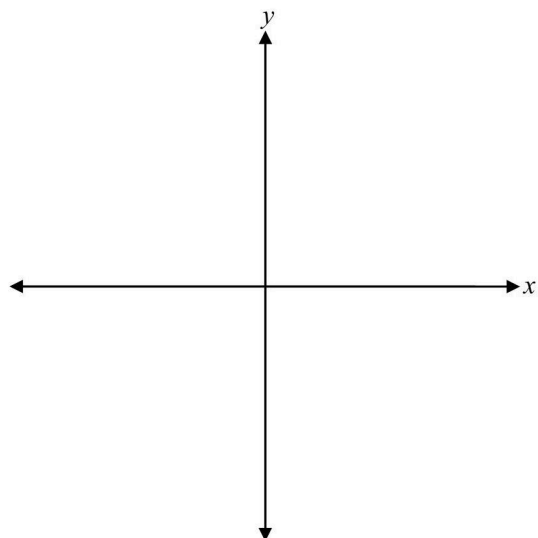
$$\frac{x-4}{3} = y \quad 1/2 \text{ mark for solving for } y$$

$$f^{-1}(x) = \frac{x-4}{3} \quad \frac{1}{2} \text{ mark for writing equation of } f^{-1}(x)$$

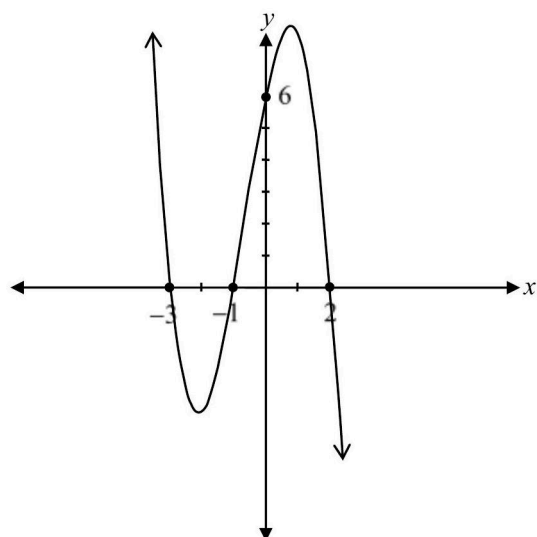
2 marks

Question 49

Sketch the graph of $P(x) = -(x+1)(x-2)(x+3)$.



Solution



1 mark for x -intercepts

$\frac{1}{2}$ mark for y -intercept

$\frac{1}{2}$ mark for end behaviour

2 marks